SCAM DETECTOR SYSTEM

Micro Project Report

Submitted to the APJ Abdul Kalam University, Kerala
for the partial fulfilment of the requirements for the award of the degree

of

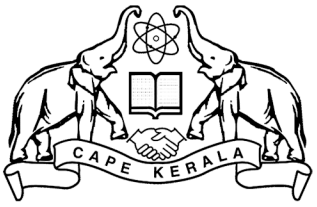
Bachelor of Technology
in
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INTRODUCTION

The rapid growth of the internet and digital communication has made online activities such as banking, shopping, messaging, and information sharing easier and more convenient. However, this growth has also led to an increase in cyber fraud and online scams. Scammers use fraudulent websites, suspicious messages, fake emails, misleading phone numbers, and malicious QR codes to deceive users and steal sensitive information such as passwords, banking details, and personal data.

The **Scam Detection System** is a Java-based desktop application developed to help users identify potentially suspicious online content. The system provides a user-friendly graphical interface through which users can analyse different types of digital content and receive information about possible security risks. It aims to improve user awareness and encourage safer internet practices.

The application includes multiple detection modules, such as website URL analysis, message analysis, email analysis, phone number analysis, screenshot analysis, and QR code analysis. These modules examine the submitted information using appropriate analysis techniques to identify suspicious patterns and potential indicators of scams. The system also incorporates database functionality to maintain scan history, allowing users to review previous analyses. Supabase is used for cloud-based database functionality and backend integration.

The application is developed using Java and the Swing framework for its graphical user interface. Supporting libraries and external services are used where required to provide additional functionality, such as QR code processing and AI-assisted image analysis.

The main objective of this project is to provide a simple and accessible tool that helps users recognize potential online scams before interacting with suspicious content. Although the system can assist in identifying possible threats, its results should be treated as guidance rather than a guarantee of security. The project demonstrates how software development, database integration, and automated analysis techniques can be combined to support cybersecurity awareness and safer digital communication.

SCOPE OF THE MICRO-PROJECT

The scope of the Scam Detection System is to develop a user-friendly desktop application that helps users identify potentially fraudulent and suspicious digital content. The project focuses on improving cybersecurity awareness by providing a convenient platform to analyse different types of information that may be associated with online scams.

The major areas covered by the project are:

1. Website URL Detection: Analyses website URLs to identify suspicious patterns and potential indicators of fraudulent websites.
2. Message Detection: Examines text messages to identify suspicious words, misleading offers, and possible scam-related content.
3. Email Detection: Analyses email content to identify potential phishing attempts and fraudulent communication.
4. Phone Number Analysis: Helps users assess phone numbers that may be associated with suspicious or fraudulent activities.
5. Screenshot Analysis: Uses image analysis techniques and AI-assisted processing, where available, to examine screenshots for possible scam indicators.
6. QR Code Detection: Allows users to analyse QR codes and examine the information or links encoded within them for potential risks.
7. Scan History Management: Maintains records of previous analyses using database functionality, including cloud-based storage through Supabase where configured.
8. User-Friendly Interface: Provides a graphical user interface developed using Java Swing, allowing users to access the different detection modules easily.

The project is intended to serve as an initial layer of assistance for identifying possible online scams. It can be further enhanced in the future by integrating updated threat intelligence databases, real-time URL reputation services, improved machine-learning models, and additional detection mechanisms. The system does not guarantee that every scam will be detected, and its results should be used as guidance rather than as a replacement for careful verification.

TECHNOLOGIES USED

1. **Java:** Used as the primary programming language to implement the application's logic and detection modules.

2. **Java Swing:** Used to develop the graphical user interface (GUI) of the application.

3. **Supabase:** Used for cloud database functionality and storing and retrieving scan history.

4. **PostgreSQL:** Used as the relational database associated with Supabase.

5. **Supabase Edge Functions:** Used to implement backend operations and server-side functionality.

6. **Llama API:** Used to support AI-assisted analysis of screenshots and images to identify potential scam indicators.

7. **ZXing Library:** Used for decoding and processing QR codes.

8. **Eclipse IDE:** Used for Java development, debugging, and testing.

9. **Git and GitHub:** Used for version control, source code management, and project collaboration.

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PROBLEM STATEMENT

The increasing use of the internet and digital communication has resulted in a rise in online scams, including phishing websites, fraudulent messages, fake emails, suspicious phone numbers, and malicious QR codes. These scams can deceive users into revealing sensitive information, making fraudulent payments, or exposing their personal data.

Many users lack the technical knowledge required to identify suspicious digital content. Furthermore, checking different types of potential scams often requires using separate tools, making the process inconvenient and time-consuming.

Therefore, there is a need for a unified and user-friendly system that helps users analyse different types of digital content and identify potential scam indicators. The proposed Scam Detection System aims to address this problem by providing multiple analysis modules within a single desktop application, supported by AI-assisted analysis and scan history management where available.

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LITERATURE REVIEW/ BACKGROUND

Online scams have become a major cybersecurity concern due to the increasing use of digital communication, online payments, and internet-based services. Common threats include phishing websites, fraudulent emails, scam messages, suspicious phone numbers, and malicious QR codes. Various techniques have been developed to identify such threats and reduce the risk of online fraud.

Traditional scam detection techniques use rule-based analysis, keyword matching, URL pattern checking, and predefined conditions to identify suspicious content. These methods are useful for detecting known scam indicators and identifying potentially harmful information. However, their effectiveness depends on the rules and patterns used, and they may not identify every new or sophisticated scam.

Artificial Intelligence (AI) provides additional capabilities for analysing text and images. AI models can examine the context of messages and screenshots to identify possible signs of fraud that may not be captured by simple rules alone. Combining traditional detection techniques with AI-assisted analysis can provide a more flexible approach to identifying potential scams.

The proposed Scam Detection System incorporates both traditional analysis techniques and AI-assisted analysis. It provides separate modules for website URLs, messages, emails, phone numbers, screenshots, and QR codes. Traditional methods are used where applicable to examine suspicious patterns and predefined indicators, while the Llama API supports AI-assisted analysis of screenshots and images. Supabase provides cloud database functionality for storing and retrieving scan history.

The system integrates these features into a single Java Swing desktop application, allowing users to analyse different types of digital content through a common interface. The project aims to improve accessibility and cybersecurity awareness by helping users recognize potential scam indicators. However, the results are intended as guidance and cannot guarantee that all fraudulent content will be detected.

IMPLEMENTATION

The Scam Detection System is implemented as a Java-based desktop application using the Swing framework. The application provides a graphical user interface through which users can access different scam detection features. It combines traditional rule-based analysis with AI-assisted image analysis to identify potential scam indicators.

1. Home Module: Provides the main interface and navigation to the various detection modules.

2. Website URL Analysis: Examines submitted URLs using predefined rules and suspicious patterns to identify possible indicators of fraudulent websites.

3. Message Analysis: Analyses message content for suspicious keywords, misleading offers, and common scam-related patterns.

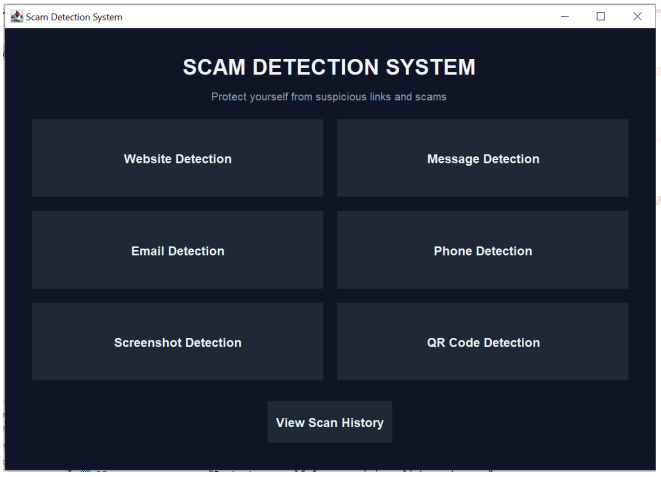
4. Email Analysis: Examines email content for potential phishing indicators and suspicious communication patterns.

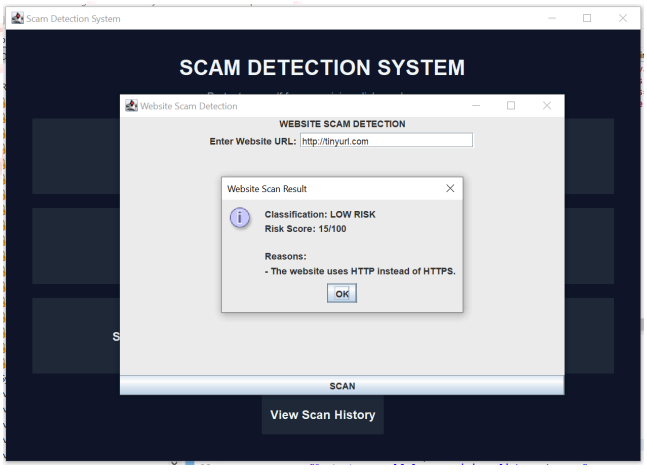
5. Phone Number Analysis: Allows users to submit phone numbers for analysis using the detection methods implemented in the application.

6. Image Analysis: Uses the Llama API for AI-assisted analysis of screenshots and images to identify possible signs of scams.

7. QR Code Analysis: Uses the ZXing library to decode QR codes and extract their encoded information for further inspection.

8. Scan History: Integrates with Supabase and its associated PostgreSQL database to store and retrieve scan records, allowing users to review previous analyses.





DETAILS OF JAVA CLASSES AND ARCHITECTURE

Details of Java Classes

The Scam Detection System is developed using Java and the Swing framework. The application follows a modular structure in which separate classes handle the graphical interface, scam analysis, AI integration, and database operations.

1. **HomeFrame.java:** Provides the main interface and navigation to all detection modules.
2. **WebsiteFrame.java:** Accepts website URLs, initiates analysis, displays results, and saves scan records.
3. **URLAnalyzer.java:** Applies predefined rules to identify suspicious URL patterns and calculate a risk score.
4. **MessageFrame.java:** Provides the interface for entering and analysing suspicious text messages.
5. **MessageAnalyzer.java:** Examines messages for suspicious keywords, links, urgent requests, and common scam indicators.
6. **EmailFrame.java:** Accepts email subjects and content for analysis and displays the results.
7. **EmailAnalyzer.java:** Checks email content for potential phishing indicators and suspicious patterns.
8. **PhoneFrame.java:** Accepts phone numbers and related information for scam assessment.
9. **PhoneAnalyzer.java:** Applies predefined rules to identify potential warning signs in phone-related information.
10. **ScreenshotFrame.java:** Allows users to select screenshots and view AI-assisted analysis results.
11. **AdditionalAnalyzer.java:** Communicates with the Groq API to perform AI-assisted screenshot analysis.
12. **QRFrame.java:** Provides the interface for selecting and analysing QR code images.
13. **QRAnalyzer.java:** Uses the ZXing library to decode QR codes and extract their encoded information.
14. **GroqAIService.java:** Communicates with the Supabase Edge Function to obtain AI-assisted analysis results for supported detection modules.
15. **DatabaseManager.java:** Manages the local SQLite database and saves scan records.
16. **HistoryFrame.java:** Retrieves and displays scan history from Supabase.
17. **ResultFrame.java:** Provides a separate graphical interface for displaying analysis results.
18. **DatabaseTest.java:** Tests local database initialization and scan-record storage.

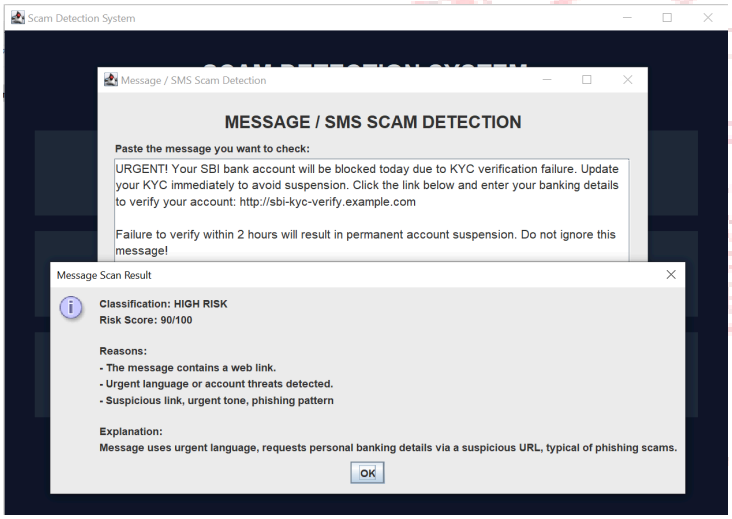
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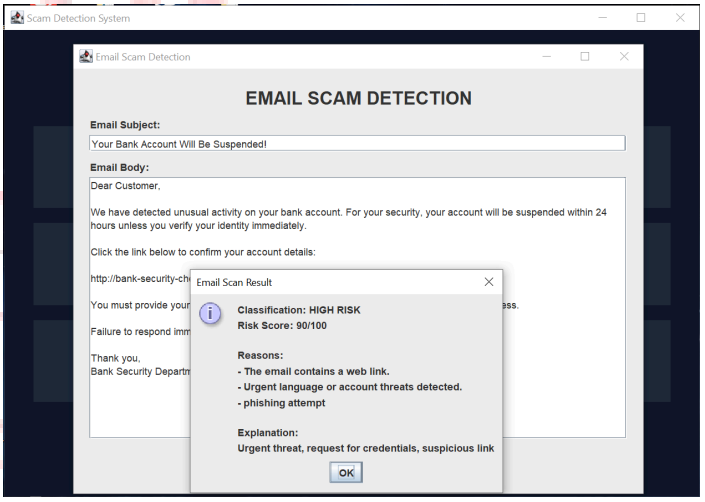
RESULTS AND DISCUSSION

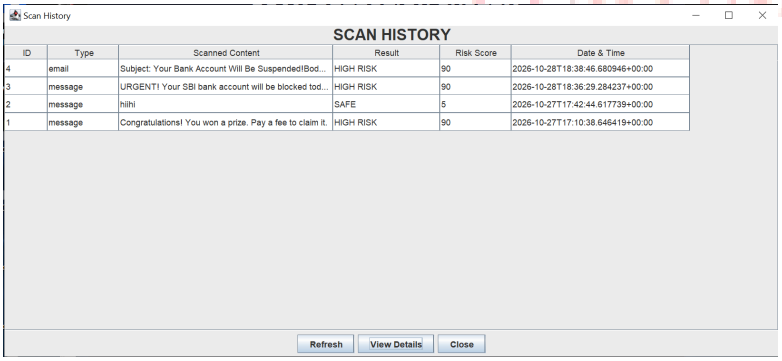
The Scam Detection System provides a unified desktop application for analysing potentially suspicious digital content. It integrates traditional rule-based techniques, AI-assisted analysis, and database functionality.

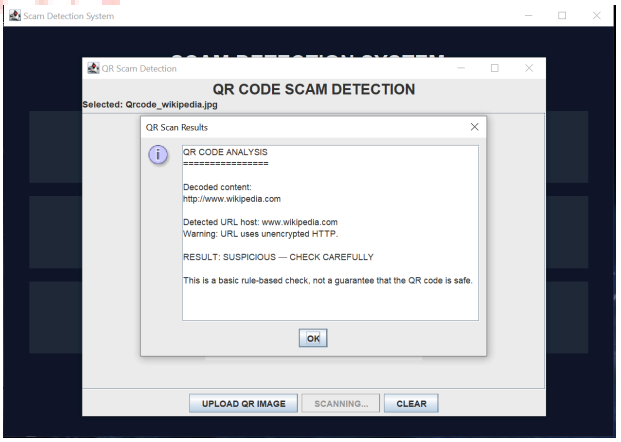
The application includes modules for website URLs, messages, emails, phone numbers, screenshots, and QR codes. It displays analysis results and supports scan-history management using local SQLite storage and Supabase cloud integration.

The modular design and user-friendly interface make the system convenient to use. However, detection results depend on the implemented rules, available information, and AI responses. The system assists users in identifying potential scam indicators but cannot guarantee complete detection.









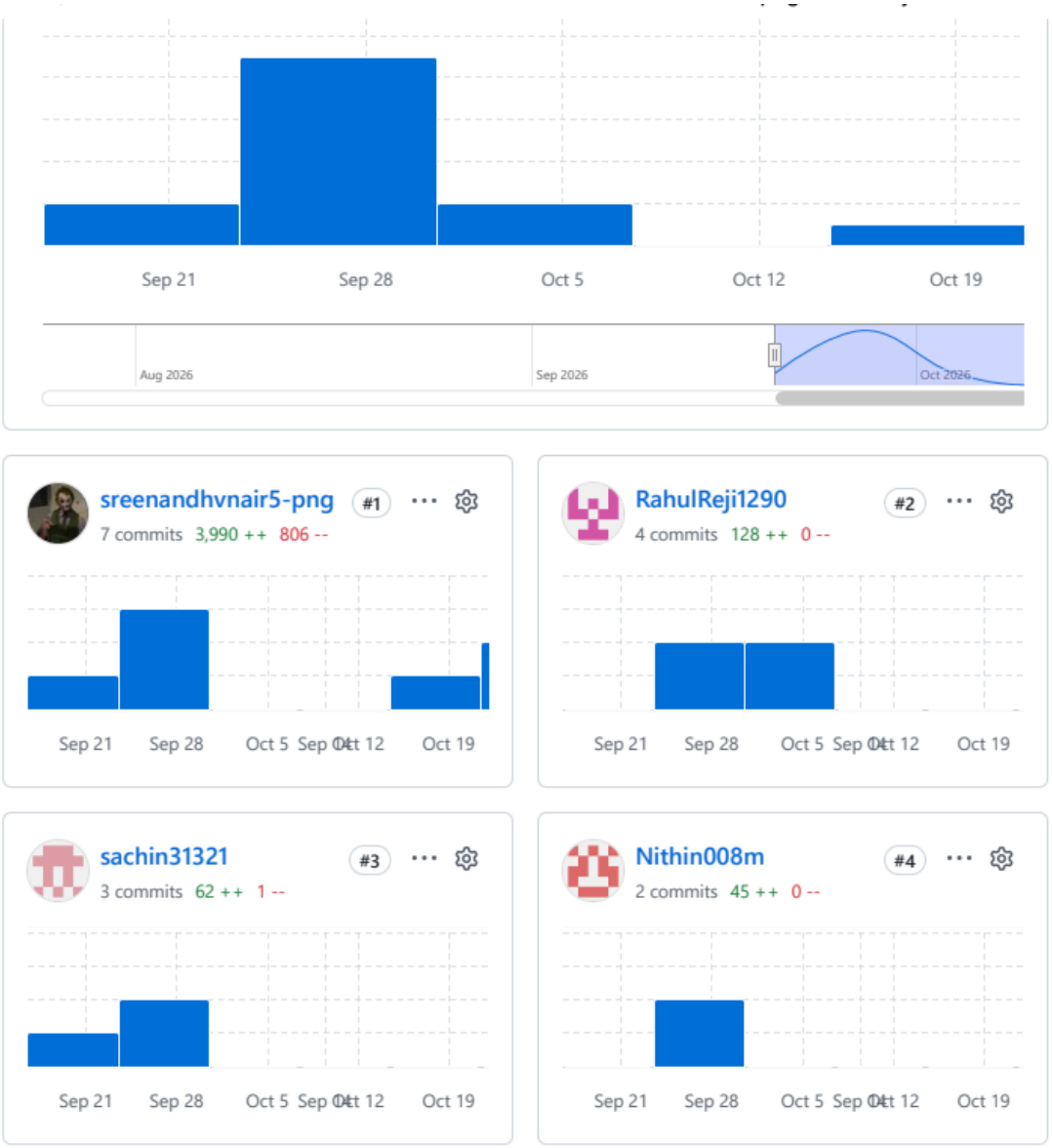
GitHub Repository Insights

GitHub is used for source code management and version control of the Scam Detection System. Repository insights provide information about development activity, commits, and contributor participation. These records help demonstrate the project's development progress and maintain a history of code changes.

GitHub Repository Link:

The source code and development history of the Scam Detection System are maintained on GitHub.

Repository: https://github.com/sreenandhvnair5-png/MicroProject





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CONCLUSION

The Scam Detection System was developed as a Java-based desktop application to help users identify potentially fraudulent digital content. It integrates traditional rule-based detection techniques with AI-assisted analysis to examine website URLs, messages, emails, phone numbers, screenshots, and QR codes.

The application provides a user-friendly interface using Java Swing and incorporates database functionality through SQLite and Supabase for scan storage and history management. The modular architecture makes the system easier to maintain and extend.

This microproject demonstrates the practical application of Java programming, database integration, API communication, and cybersecurity techniques. The system can be further improved by integrating updated threat intelligence sources, enhancing detection accuracy, and expanding its analysis capabilities.

Overall, the project aims to improve cybersecurity awareness and help users make more informed decisions when encountering potentially suspicious online content. The results should be treated as guidance, as no detection system can guarantee the identification of every scam.

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